

latest architectures (Intel's Core 2 Duo and AMD's Athlon 64) are significantly faster than previous-generation products, with significant performance leads across most applications. Let's take a side-by-side look (literally) at all the features the latest architectures from these two CPU giants sport.

The Philosophy Of The Core...

Intel's Core range of CPUs—"Core" being the architectural handle for the Core 2 Duos—are being made on a 65 nm fab. Intel is no stranger to 65 nm: their Pentium D Presler series were the first 65 nm CPUs, remember? The earlier 90 nm core had a bad history as far as heat dissipation goes, but the 90 nm process wasn't the only culprit. Hand-in-glove were the longer pipelines the Prescott sported, not to mention that it sported more of them than did the older Northwood cores.

The die shrink and the 65 nm fab collectively translate to reduced power leakage—less heat dissipation. Besides this, the Core architecture also has fewer transistors than the previous Presler (the 9xx series) cores. Not only this, the size of the transistors has also come down, so while the Core 2 Duo's transistor count is less than that of the Pentium D 9xx family, the density of the transistors can be greater.

Intel is talking about "performance per watt" with the Core range. Gone are the gigahertz wars where 4 GHz was considered the Holy Grail. These days the focus is on

controller (namely DDR2 support), the AM2 Socket-based AMD 64s are nearly identical to their 939-pin brethren. AMD is still using Fab 30, and a newer Fab 36—both 90 nm fabs, at their plant in Dresden, Germany.

AMD has followed "true" dual-core architecture from the very start with their 939 dual-cores. The cores can communicate with each other through the Crossbar Controller (internal to the processor). In the case of Intel dual-cores, any communication between the cores occurs via the FSB (external to the processor), which causes latency, especially when you consider the speed of the FSB is a measly 800 MHz. Compare this number to the CPU clock figures and you'll get the drift.

AMD's On-Die Memory Controller

AMD 64's mainstay has been the Integrated Memory Controller or IMC. This has been one of the major advancements in desktop processing in modern times. What is it? Simply take the memory controller traditionally present on the MCH (Memory Controller Hub) or Northbridge, and relocate it on the CPU die. What happens? Well, for one, the pin count increases (939 and now 940 are a lot of pins!). Performance-wise, the bottleneck between the CPU and the memory is removed, because the FSB route between CPU and memory becomes defunct. Communication occurs at CPU clock speeds rather than FSB clock speeds. Communi-

Prefetching is the name of the cache game. The Core 2 Duo features two data and one instruction prefetcher per core, and two prefetchers for the L2 cache, making for a grand total of eight prefetchers

performance with power saving, and Intel has learnt their lesson from their 130 W Prescott cores, dubbed "nuclear reactors" by many—that hot doesn't pay, at least as far as TDP (Thermal Design Power) goes!

Firstly, the Core 2 Duo loses quite a few pipelines. Of course, clock frequencies will be lower due to fewer pipelines, but the Core micro-architecture has some other nifty tricks up its sleeve. The Core 2 Duos are designed from the ground up to be dual-core solutions, unlike the Pentium D CPUs, which were "pseudo" dual-cores—more aptly "double cores." The Smithfield and Presler processors (Pentium Ds) were basically double cores—the Smithfield consisted of two Prescott cores, while the Presler comprised two Cedar Mill cores.

Looking at the Conroe's Core architecture, it's easy to determine its roots: the quad-bus has been lifted right out of the NetBurst architecture, while the power-saving features smack strongly of the Pentium M Dothan and Banias processors.

Memory Subsystems—Advanced Smart Cache

Minimising the role of memory latencies as a performance deterrent was a major objective for Intel. Also, the processor cache subsystem had to be beefed up in order to counter the latencies that occur due to the lack of an on-die memory controller, as on the AMD64. The L1 cache is a hefty 128 KB (64

AMD—Latency Vs. Bandwidth

Due to the IMC, and the AMD's very architecture, their processors are more sensitive to changes in memory timings rather than bandwidth (read as frequency) changes. This is why AMD 64 939s initially performed much better than the AM2 equivalents; the latencies associated with DDR 2 were much higher.

cation with the memory therefore becomes much faster, and in comparison to all Intel CPUs, AMD 64s utilise memory bandwidth more efficiently.

Hand-in-hand with the IMC, AMD 64s have a massive L1 cache size advantage. 128 KB of L1 cache per core, is a huge figure: it contributes significantly to the percentage efficiency of memory utilisation. The L1 and L2 cache remains identical to Socket 939 processors however, and the L2 cache is either 512 KB or 1 MB depending on the model.

Execution Subsystem And Decoding: Complex Decoders Galore

Let's take a very brief look at what a decoder on a CPU is supposed to do. CPUs are bombarded with instructions,



2000
Pentium 4—in its earliest avatar, NetBurst was hot, P6 was not!

2001

Athlon XPs, faster than the Athlon models, outperform the initial Pentium 4s



2002

Northwood cores emerge, featuring HT and higher clock speeds



2003

AMD crashes Intel's party with their fearsome AMD 64s



2004
(AMD 64 continues)



... And these enjoy nearly three years of supremacy

2005

AMD's Dual Core X2s emerge—the Manchester and Toledo cores



2006

Intel retorts—the Core 2 Duo topples AMD's apple-cart

